

Performance Metrics

CSC3034 Computational Intelligence

Dr Tang Tiong Yew

Lecture outline

Introduction

Gold Standard

Percent Correct

Errors

Average Sum-Squared Error

Absolute Error

Normalised Error

Mann-Whitney U Test

Matrices

Contingency Matrices

Confusion Matrices

Receiver Operating Characteristic (ROC) Curves

Chi-Square Test

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- ▶ For example in supervised neural network trainings, the data is provided in the form of input patterns and expected output patterns. The data of expected output patterns is the gold standard.
- ▶ Gold standard is used to compare with the output of a system to evaluate the performance of the system.

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Percent correct

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Suppose we have a group of 100 stocks. A computational intelligence tool is used to predict whether the stock will outperform or perform less well than the Dow Jones average.

Last month, 90 percent of the stocks that outperform were predicted accurately, and 60 percent of the stocks that perform less well were predicted correctly.

This month, 85 percent of the stocks that outperform were predicted accurately, and 55 percent of the stocks that perform less well were predicted correctly.

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Last month, 90 percent of the stocks that outperform were predicted accurately, and 60 percent of the stocks that perform less well were predicted correctly.

This month, 85 percent of the stocks that outperform were predicted accurately, and 55 percent of the stocks that perform less well were predicted correctly.

The percent correct decreased, however, the overall ability to predict could have improved.

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and 70 percent of the stocks this month outperformed the Dow Jones average,

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The overall predictive accuracy increased, even though the predictive accuracy on the individual metrics decreased.

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- ▶ we might want to consider more than one performance metric in evaluating a system.

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This manipulation of percent correct metric introduces subjectivity, but improve its usefulness in practical sense.

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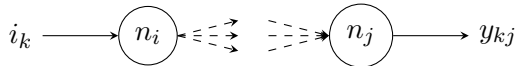
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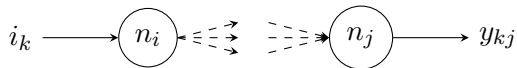
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$$\text{total error, } E_t = 0.5 \sum_{k=1}^m \sum_{j=1}^q (b_{kj} - y_{kj})^2$$

$$\text{average sum-squared error, } e = \frac{E_t}{m}$$

where m is the total number of patterns
 q is the total number of output neurons

Average sum-squared error



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Therefore it is important to be aware of how the error term is calculated when you are using average sum-squared error to compare networks.

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$$e \text{ per output neuron} = \frac{\text{average sum-squared error, } e}{\text{number of output neurons, } q}$$

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Mean absolute error

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Maximum absolute error

$$E_{\max} = \max_{k,j} |b_{kj} - y_{kj}|$$

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$$\text{normalised error, } E_n = \frac{E_t}{E_{\text{mean}}}$$

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Overall it is believed that this error measure may be the most useful one for back-propagation neural network implementations.

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To overcome this situation, it is suggested to remove the output neurons whose expected output does not change value.

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- ▶ The test evaluates whether the medians of two samples of data are the same using hypothesis testing.
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- ▶ The test will inform us to reject or not to reject the null hypothesis.

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2. to determine if the null hypothesis should be rejected.

Mann-Whitney U test

Calculation of the U value:

Method 1 quick and direct

Method 2 formula

Mann-Whitney U test

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5 runs are made with configuration A and obtain the best fitness values of

0.079 0.062 0.073 0.047 0.085

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- ▶ we are using an evolutionary algorithm with two configurations A and B, where B is our “baseline” configuration.

5 runs are made with configuration A and obtain the best fitness values of

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and 4 runs are made with configuration B and obtain the best fitness values of

0.102 0.069 0.055 0.049

n_1 and n_2 are number of runs made with configurations A and B respectively.

$$\therefore n_1 = 5, n_2 = 4$$

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Method 1: quick and direct

1. count the number of A entries that are better than each of the B entries.

	1 st B	2 nd B	3 rd B	4 th B
# of A better than				

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# of A better than	1			

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# of A better than	1	1	2	

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	1 st B	2 nd B	3 rd B	4 th B
# of A better than	1	1	2	5

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$$U_1 = \text{Total} = 9$$

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2. count the number of B entries that are better than each of the A entries.

	1 st A	2 nd A	3 rd A	4 th A	5 th A
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	1 st A	2 nd A	3 rd A	4 th A	5 th A	$U_2 = \text{Total} = 11$
# of B better than	0	2	3	3	3	

Mann-Whitney U test

Next, we arrange the measurements in the order of fitness with better fitness on the left.

0.047	0.049	0.055	0.062	0.069	0.073	0.079	0.085	0.102
A	B	B	A	B	A	A	A	B

Method 1: quick and direct

1. count the number of A entries that are better than each of the B entries.

	1 st B	2 nd B	3 rd B	4 th B	
# of A better than	1	1	2	5	$U_1 = \text{Total} = 9$

2. count the number of B entries that are better than each of the A entries.

	1 st A	2 nd A	3 rd A	4 th A	5 th A	
# of B better than	0	2	3	3	3	$U_2 = \text{Total} = 11$

3. The smaller of U_1 and U_2 is the U value, i.e. 9.

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* Note that the two values add up to $n_1 n_2$.

Mann-Whitney U test

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Method 2: formula

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1. label the rank from 1 to 9 going from left to right.
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Method 2: formula

1. label the rank from 1 to 9 going from left to right.
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$$R_1 = 1 + 4 + 6 + 7 + 8 = 26$$

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$$R_1 = 1 + 4 + 6 + 7 + 8 = 26$$

$$U_1 = n_1 n_2 + \frac{n_1(n_1 + 1)}{2} - R_1$$

Mann-Whitney U test

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Mann-Whitney U test

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Mann-Whitney U test

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For example,

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---	----	---	----	---	----	---	----	---	----

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---	----	---	----	---	----	---	----	---	----

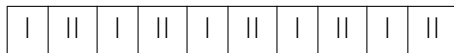
If the medians of the two samples differ a lot, the two samples will be separated clearly on the rank of fitness. U_1 and U_2 will be very different and the U value (the smaller value) will be small.

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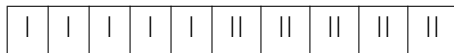
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This is the critical value that we will obtain from the Mann-Whitney U table, below which we will reject the null hypothesis.

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1. identify the critical value, and
2. if the U value is less than the critical value, reject the null hypothesis, else, we do not reject the null hypothesis.

Mann-Whitney U test

The critical values of the Mann-Whitney U for small values of n_1 and n_2 at significance level of 0.05 are given in the following table:

n_1	n_2							
	3	4	5	6	7	8	9	10
3	0	0	1	2	2	3	4	4
4	0	1	2	3	4	5	6	7
5	1	2	4	5	6	8	9	11
6	2	3	5	7	8	10	12	14
7	2	4	6	8	11	13	15	17
8	3	5	8	10	13	15	18	20
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For $n_1 = 5$ and $n_2 = 4$, the critical value is 2.

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Mann-Whitney U test

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Let say we test another configuration, which we named as C. We make five runs using configuration C and we obtain the fitness values of 0.060, 0.045, 0.043, 0.038, and 0.034. Compare with the fitness values of the four baseline cases (B), 0.102, 0.069, 0.055, and 0.049, what is the U value? Is the new configuration C statistically better than the baseline configuration B at 0.05 level of significance?

Mann-Whitney U test

Notes:

Mann-Whitney U test

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Absolute Error

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Contingency Matrices

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		Positive	Negative
Gold Standard Diagnosis	Positive	true positive (TP)	false negative (FN)
	Negative	false positive (FP)	true negative (TN)

Contingency Matrix

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Ratios:

Contingency Matrix

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- accuracy

$$\frac{TP + TN}{TP + TN + FP + FN}$$

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- ▶ false positive ratio/false alarm rate

$$\frac{FP}{(FP + TN)}$$

- ▶ true negative ratio/specificity

$$\frac{TN}{(FP + TN)}$$

- ▶ false positive ratio = 1 - specificity

Contingency Matrix

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		System outputs		
		A	B	C
Gold standard	A			
	B			
	C			

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		System outputs		
		A	B	C
Gold standard	A	40	8	2
	B	6	42	2
	C	1	1	48

Confusion matrices

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		A	B	C
Gold standard	A	0.80	0.16	0.04
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Each row should add up to 1. This is the probability to be diagnosed as a certain class given that the case belongs to a specific diagnose.

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The diagonal values represent the recall of each class.

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Each element is now the probability of that outcome. The sum of all elements is now equal to 1.0.

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	B	0.036	0.252	0.012
	C	0.002	0.002	0.096

$$\text{accuracy} = 0.480 + 0.252 + 0.096 = 0.828$$

The **accuracy** of the system is calculated by calculating the sum of the main diagonal values.

		System outputs		
		A	B	C
Gold standard	A	0.480	0.096	0.024
	B	0.036	0.252	0.012
	C	0.002	0.002	0.096

$$\text{accuracy} = 0.480 + 0.252 + 0.096 = 0.828$$

i.e. 82.8% of the diagnoses the system makes is correct.

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Therefore, we need to construct the cost matrix to investigate the cost of using the system.

Confusion matrices

Suppose that,

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The medical diagnosis B is a more serious medical problem that requires aggressive treatment with a prescription antibiotic that costs \$100.

Condition C is a very serious condition that requires hospitalisation and surgery, with a total cost of \$5000. If condition C is not diagnosed right now, the patient has only a 20-percent chance of survival. Even though it is impossible to assign a dollar value to a human life, let's say that the average insurance policy of these patients is \$100,000. That means misdiagnosed condition C will result in an average cost of \$80,000 (80% chance of death $\times$ \$100,000).

It is often assumed that correct classification/diagnosis imposes zero cost, but it is clear that in our case, the cost for a correct diagnosis is not zero.

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Now we are ready to identify the cost for each misdiagnosis and correct diagnosis based on the previous slide.

Cost matrix

Cost matrix

		System outputs		
		A	B	C
Gold standard	A	\$10	\$110	\$5,010
	B	\$110	\$100	\$5,100
	C	\$80,010	\$80,100	\$5,000

Cost matrix

		System outputs		
		A	B	C
Gold standard	A	\$10	\$110	\$5,010
	B	\$110	\$100	\$5,100
	C	\$80,010	\$80,100	\$5,000

Then we multiply this cost matrix with the final confusion matrix element by element to obtain the **final cost matrix**.

Final cost matrix

Final cost matrix

		System outputs		
		A	B	C
Gold standard	A	4.8	10.56	120.24
	B	3.96	25.2	61.2
	C	160.02	160.20	480.00

Final cost matrix

		System outputs		
		A	B	C
Gold standard	A	4.8	10.56	120.24
	B	3.96	25.2	61.2
	C	160.02	160.20	480.00

The average estimated cost of each application of the diagnostic system is given by the sum of all elements in the final cost matrix, which is \$1,026.18.

The cost calculations from confusion matrices can be used to fine-tune a computational intelligence system.

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For example a neural network-based diagnostic system is first trained. Then the threshold levels of the output neurons can be adjusted by calculating the costs associated to choose the combination that produces the lowest system cost.

The cost calculations from confusion matrices can be used to fine-tune a computational intelligence system.

For example a neural network-based diagnostic system is first trained. Then the threshold levels of the output neurons can be adjusted by calculating the costs associated to choose the combination that produces the lowest system cost.

Even if cost information is not available, the system can still be tuned to minimise the off-diagonal numbers, of which the sum represents the percent incorrect.

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- ▶ An ROC curve corresponds to a system configuration.

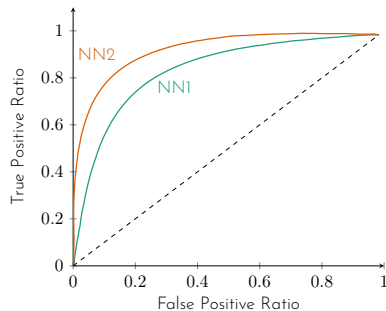
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- ▶ An ROC curve is a plot of the true positive ratio (sensitivity) versus the false positive ratio ($1 - \text{specificity}$).
- ▶ It is typical to plot the points for different values of output threshold, and connect them with line segments or a smooth curve to form the ROC curve.

Hypothetically the ROC curves for two neural networks are generated as follows:

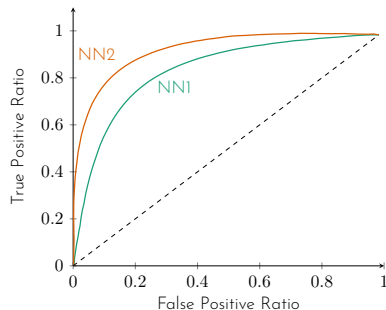
ROC curves

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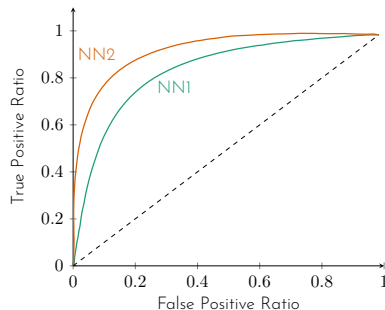
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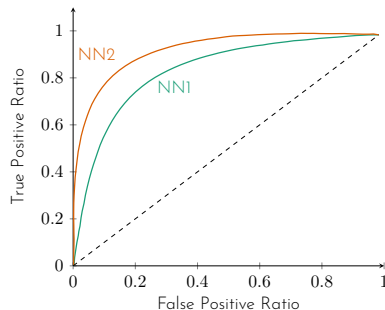
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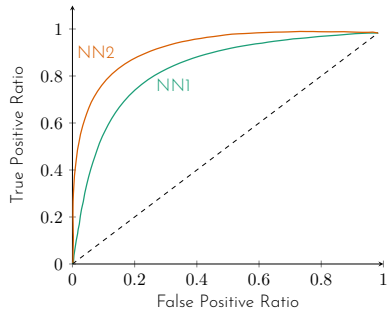
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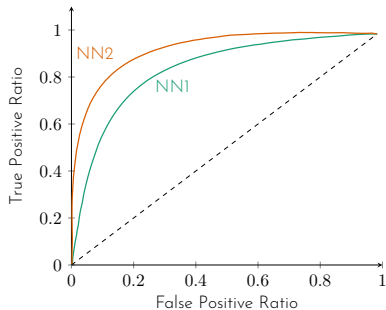
The questions of concern are:

- ▶ How do we interpret the ROC curves?
- ▶ Which is the neural network with better performance?

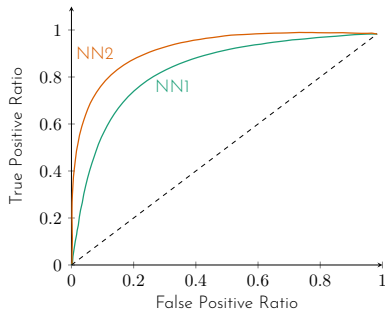
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ROC curves

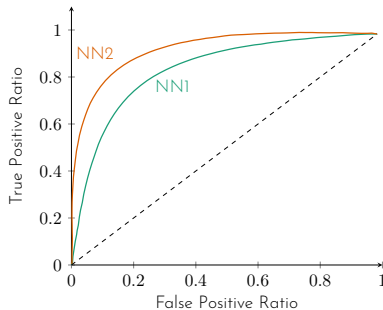


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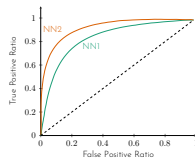
- It always lies above the major diagonal.



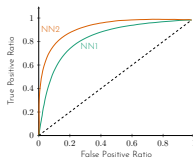
An ROC curve has two attributes:

- ▶ It always lies above the major diagonal.
- ▶ It is always monotonically increasing in value from left to right.

ROC curves

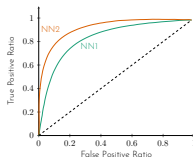


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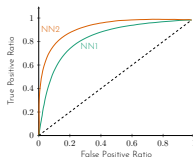


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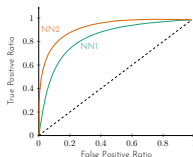
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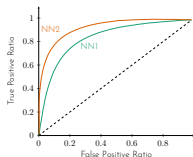
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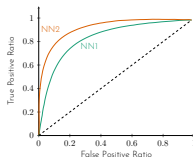
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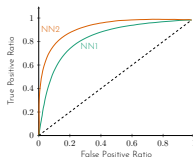
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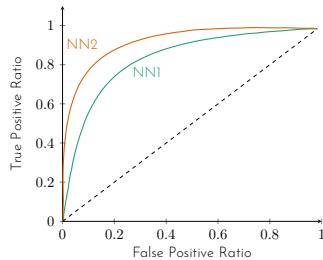


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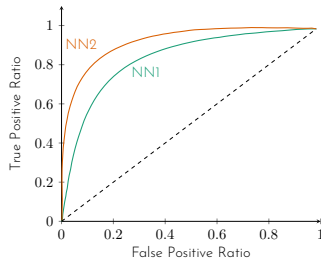


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 - generate a smooth curve through the points and calculate the area under it, or
 - join the points with straight lines and calculate the area using trapezoidal rule.

ROC curves



ROC curves



The shape of the curve indicates the sensitivity of the system performance to the threshold value.

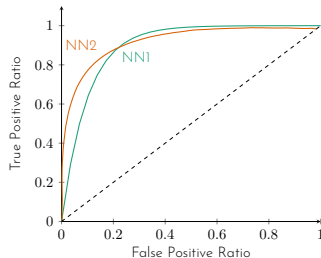
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Chi-square (χ^2) test is used to evaluate the performance of the system where

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To use chi-square test, the categories must be **mutually exclusive** and **collectively exhaustive**. The data is assumed to be normally distributed.

Chi-square test

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1. determine the chi-square value χ^2 ,
2. determine the degree of freedom df ,
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4. determine if the null hypothesis (the expected frequency distribution and the observed frequency distribution has no significant difference) should be rejected.

Chi-square test

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$$\chi_1^2 = \frac{(6 - 5)^2}{5} + \frac{(10 - 10)^2}{10} + \frac{(14 - 15)^2}{15} + \frac{(20 - 20)^2}{20} = 0.267$$

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$$\chi_2^2 = \frac{(2 - 5)^2}{5} + \frac{(15 - 10)^2}{10} + \frac{(8 - 15)^2}{15} + \frac{(25 - 20)^2}{20} = 8.82$$

Chi-square test

To determine the degree of freedom df

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In our example, there are 4 categories, and therefore

$$df = 4 - 1 = 3$$

Chi-square test

To obtain critical values from the chi-square distribution table

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df	Area to the right									
	0.995	0.990	0.975	0.950	0.900	0.100	0.050	0.025	0.010	0.005
1	0.000	0.000	0.001	0.004	0.016	2.706	3.841	5.024	6.635	7.879
2	0.010	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210	10.597
3	0.072	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.345	12.838
4	0.207	0.297	0.484	0.711	1.064	7.779	9.488	11.143	13.277	14.860
5	0.412	0.554	0.831	1.145	1.610	9.236	11.070	12.833	15.086	16.750
6	0.676	0.872	1.237	1.635	2.204	10.645	12.592	14.449	16.812	18.548
7	0.989	1.239	1.690	2.167	2.833	12.017	14.067	16.013	18.475	20.278
8	1.344	1.646	2.180	2.733	3.490	13.362	15.507	17.535	20.090	21.955
9	1.735	2.088	2.700	3.325	4.168	14.684	16.919	19.023	21.666	23.589
10	2.156	2.558	3.247	3.940	4.865	15.987	18.307	20.483	23.209	25.188
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮

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8	1.344	1.646	2.180	2.733	3.490	13.362	15.507	17.535	20.090	21.955
9	1.735	2.088	2.700	3.325	4.168	14.684	16.919	19.023	21.666	23.589
10	2.156	2.558	3.247	3.940	4.865	15.987	18.307	20.483	23.209	25.188
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⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮

For degree of freedom $df = 3$, at 0.05 level of significance, our critical value is 7.815.

Chi-square test

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If χ^2 is sufficiently big, it means that the expected and observed frequency distribution are different.

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If χ^2 is sufficiently big, it means that the expected and observed frequency distribution are different.

The boundary or threshold value for χ^2 for us to conclude that the difference between the two distributions is significant is the critical value.

Chi-square test

Therefore,

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If the χ^2 value is greater than or equal to the critical value,

we **reject** the null hypothesis,

Chi-square test

Therefore,

If the χ^2 value is greater than or equal to the critical value,

we **reject** the null hypothesis,

whereas if the χ^2 value is less than the critical value,

we **do not reject** the null hypothesis.

Chi-square test

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$$\chi_2^2 = 8.82 \geq 7.815,$$

the null hypothesis is rejected at 0.05 significance level for second configuration.

Chi-square test

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$$\chi_1^2 = 0.267 < 7.815,$$

the null hypothesis is not rejected at 0.05 significance level for first configuration.

This means that the difference between the expected frequency distribution and the observed frequency distribution in the first configuration is not significant.

$$\chi^2_2 = 8.82 \geq 7.815,$$

the null hypothesis is rejected at 0.05 significance level for second configuration.

$$\chi^2_2 = 8.82 \geq 7.815,$$

the null hypothesis is rejected at 0.05 significance level for second configuration.

This means that the expected frequency distribution and the observed frequency distribution in the second configuration are significantly different.

Chi-square test

Therefore, we would say that the first system configuration is better than the second one.

Therefore, we would say that the first system configuration is better than the second one.

Or at the very least, the first system configuration is acceptable while the second one is not.

Chi-square test

Notes:

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- ▶ Chi-square test evaluates the performance of the entire system at once, as oppose to ROC curve, which is designed to analyse one output neuron (one category) at a time.

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- ▶ Chi-square test evaluates the performance of the entire system at once, as oppose to ROC curve, which is designed to analyse one output neuron (one category) at a time.
- ▶ It is necessary to determine the categories that are possible. In some cases, not all categories are expected. The categories with zero expected frequency have to be handled with care, whether to remove them, or to include them and perform division by zero.

Chi-square test

Notes:

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- ▶ In general, chi-square test should only be used when $\leq 20\%$ of the expected counts are < 5 , and all individual counts are at least 1.

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- ▶ In general, chi-square test should only be used when $\leq 20\%$ of the expected counts are < 5 , and all individual counts are at least 1.
- ▶ It should be obvious that when designing or tuning a computational intelligence system, the goal is to minimise the chi-square value, i.e. the observed frequency distribution should be as similar to the expected frequency distribution as possible.

A classification system categorises objects into 3 categories, A, B, and C. A test set consists of 20 items for each class is used to perform testing on two classification algorithms. The result of algorithm I shows 30 A, 15 B, and 15 C whereas algorithm II shows 23 A, 18 B, and 19 C. Use χ^2 test to evaluate the two algorithms.

Introduction

Gold Standard

Percent Correct

Errors

Average Sum-Squared Error

Absolute Error

Normalised Error

Mann-Whitney U Test

Matrices

Contingency Matrices

Confusion Matrices

Receiver Operating Characteristic (ROC) Curves

Chi-Square Test

A large, curved banner with the Sunway University logo and name. The logo features a stylized sun and the text "SUNWAY UNIVERSITY". The banner is supported by several metal poles and is positioned in front of a modern building with a glass facade. The background shows a clear blue sky and some greenery.

SUNWAY
UNIVERSITY

Thank you

CSC3034 Current Course Update

10 Performance Metrics - presentation

Synchronized from the current CSC3034 course site on 14 August 2026.

- Current labs connect core CI algorithms with physical-AI simulation examples.
- Isaac Sim examples use its bundled Python 3.12 environment and current launchers.
- Every simulator-dependent example includes a CPU/Matplotlib learning fallback.

Current materials author: Dr Tang Tiong Yew

See the synchronized lab notes and runnable examples for full instructions.